# DEPARTMENT OF ELECTRONICS AND INSTRUMENTATION

# TOPIC: Design and Implementation of a Radio-Controlled Fixed-Wing Aircraft Using BLDC Propulsion and Servo-Actuated Control Surfaces


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# ABSTRACT

The project focuses on the **design, fabrication, and control of a fixed-wing radio-controlled (RC) aircraft** employing a **brushless DC (BLDC) motor propulsion system** and **servo-actuated aerodynamic control surfaces**. The system is powered by a **3-cell LiPo battery** and controlled wirelessly through an **RF transmitter-receiver pair**, enabling real-time control of the aircraft's motion and stability.
The **primary propulsion** is achieved using a high-efficiency **outrunner BLDC motor**, which converts electrical energy into thrust via a two-blade propeller. The **Electronic Speed Controller (ESC)** regulates the motor's speed by decoding throttle signals from the RF receiver and adjusting the duty cycle of pulse-width modulation (PWM) signals. The ESC also incorporates a **Battery Eliminator Circuit (BEC)** that supplies a stable 5V power line to the receiver and servomotors, eliminating the need for additional power regulation circuitry.
Three **9g micro servo motors** are deployed to actuate the aircraft's **ailerons**, **elevator**, and **rudder**, which provide control over roll, pitch, and yaw respectively. Each servo receives a dedicated PWM signal from the RF receiver, allowing precise deflection of control surfaces. The ailerons enable **banking maneuvers**, the elevator governs **climb and descent**, while the rudder manages **directional stability**. This arrangement replicates the fundamental aerodynamic control principles used in full-scale aircraft.
The airframe is constructed using **lightweight thermocol**, chosen for its high lift-to-weight ratio, structural simplicity, and ease of modification. The combination of low mass and efficient electronic control enables stable flight at moderate power consumption, making the system ideal for academic research, aerodynamic testing, and UAV prototype development.
The project effectively integrates **mechanical design, power electronics, control systems, and wireless communication** to realize a compact, fully functional aerial vehicle. This work serves as a foundation for future enhancements such as the integration of **autonomous flight controllers**, **GPS-based navigation**, and **sensor fusion techniques** for stabilization and path planning.

# ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our project, **"Design and Development of an RC Airplane using Servo Motors, RF Module, BLDC Motor, and ESC."**

First and foremost, we extend our sincere thanks to our project supervisor, **[Faculty Name]**, Department of **Electronics and Instrumentation Engineering, NIT Agartala**, for their valuable guidance, encouragement, and continuous support during the course of this project. Their insights and feedback were instrumental in refining our work and helping us understand the intricate balance between electronics and aerodynamics in unmanned systems.

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# INTRODUCTION

The rapid advancement of electronics and wireless communication has significantly influenced the evolution of unmanned aerial systems (UAS) and remote-controlled (RC) aircraft. RC airplanes serve as an excellent educational platform to understand the interplay of aerodynamics, control systems, propulsion, and communication technologies. This project, titled **"Design and Development of an RC Airplane using Servo Motors, RF Module, BLDC Motor, and ESC,"** aims to design, build, and demonstrate a lightweight and efficient fixed-wing aircraft capable of remote-controlled flight using radio frequency (RF) communication.

An RC airplane is essentially a miniature version of a full-scale aircraft that can be controlled remotely from the ground by an operator using a transmitter. It typically consists of a **propulsion system (motor and propeller)**, **aerodynamic control surfaces (ailerons, elevator, and rudder)**, and an **electronic control system (receiver, servos, ESC, and power supply)**. The transmitter sends control signals via RF communication, which are received by the onboard receiver. These signals are then processed to actuate various components, such as the servos and motor, to achieve controlled flight.

The heart of the propulsion system is the **Brushless DC (BLDC) motor**, chosen for its high efficiency, low weight, and reliability. The motor speed is regulated using an **Electronic Speed Controller (ESC)**, which interprets throttle signals from the receiver and adjusts the motor's power accordingly. The **LiPo (Lithium Polymer) battery** provides the required power supply, offering high energy density and discharge rates suitable for flight applications.

For flight control, the model employs **micro servo motors** connected to the **ailerons, elevator, and rudder**. These control surfaces enable three-axis maneuverability—roll, pitch, and yaw—allowing the aircraft to perform banking turns, ascend, descend, and change direction. The servos are lightweight yet powerful enough to move the aerodynamic surfaces against air resistance. Each servo is connected to a dedicated channel on the RF receiver, ensuring smooth and responsive control.

The **RF transmitter–receiver module** operates at **2.4 GHz**, which provides stable, long-range, and interference-free communication. The transmitter converts joystick movements into PWM (Pulse Width Modulation) signals that are wirelessly sent to the receiver onboard the aircraft. The receiver, in turn, distributes these signals to the

corresponding components—the ESC for throttle control and the servos for aerodynamic adjustments.

The **airframe** of the RC airplane is constructed from lightweight materials such as **thermocol (foam board)** to ensure adequate lift generation with minimal thrust requirements. The structural design ensures proper weight distribution, aerodynamic stability, and sufficient space for component integration. The wings provide the lift, while the tail assembly ensures stability and control during flight.

Through this project, students gain a comprehensive understanding of **aerodynamics, electronics, control systems, and wireless communication principles**. It bridges theoretical knowledge and practical implementation by integrating components such as the BLDC motor, ESC, servos, and RF modules into a functional aircraft system. Furthermore, the project lays the foundation for advanced UAV research, such as autonomous flight, GPS-based navigation, and sensor-assisted stability systems.

In summary, this project demonstrates the successful integration of mechanical design, electrical systems, and communication protocols to achieve stable flight control. The design choices—such as using a BLDC motor for propulsion, servo motors for control, and a 2.4 GHz RF module for communication—were made based on performance efficiency, reliability, and compatibility with lightweight aircraft systems. This project not only provides a strong conceptual and practical understanding of RC aircraft design but also opens avenues for future research and innovation in unmanned aerial vehicle technology.

## SCOPE OF THE PROJECT:-

The **scope of this project** encompasses the complete design, development, testing, and functional demonstration of a **radio-controlled (RC) airplane** that can be manually operated using a **2.4 GHz RF transmitter and receiver system**. The project integrates key domains such as **aerodynamics, embedded electronics, propulsion systems, and control mechanisms**, providing a comprehensive understanding of how various subsystems work together to achieve stable and responsive flight.

The main objective of this project extends beyond merely constructing a flying model—it is intended to serve as an **educational and experimental platform** for students and researchers interested in unmanned aerial systems (UAS), flight control systems, and IoT-based aerial applications.

***The project's scope includes:***

1. **Design and Fabrication of Airframe:**
    - Designing a lightweight, aerodynamically stable airframe using materials like thermocol or Depron foam.
    - Optimizing the shape, wingspan, and control surfaces to ensure efficient lift generation and balance.
2. **Integration of Propulsion and Control Systems:**
    - Installing and configuring a **BLDC motor** for thrust generation, controlled via an **Electronic Speed Controller (ESC)**.
    - Implementing **servo motors** to operate the ailerons, rudder, and elevator for full 3-axis control (roll, pitch, yaw).
3. **Wireless Communication and Control:**
    - Using an **RF transmitter–receiver pair (2.4 GHz)** for reliable long-range signal transmission.
    - Ensuring real-time response between pilot commands and aircraft motion.
4. **Power Management:**
    - Incorporating a **LiPo battery** with sufficient discharge capacity to power the ESC, motor, and servos.
    - Maintaining optimal weight-to-power ratio for longer flight duration and efficiency.
5. **Testing and Flight Evaluation:**
    - Performing ground and flight tests to assess the aircraft's stability, maneuverability, and control accuracy.
    - Fine-tuning control surfaces and throttle response for optimal performance.
6. **Educational and Research Application:**
    - Serving as a foundation for advanced projects like **autonomous drones**, **GPS-guided navigation**, or **sensor-based flight stabilization systems**.
    - Providing a hands-on learning experience in fields such as **control theory, mechatronics, embedded systems, and wireless communication**.

***Future Expansion:***

The project can be expanded to include:

- **Autonomous flight control** using a flight controller like **Pixhawk or ArduPilot**.

- Integration of **GPS, gyroscope, accelerometer, and barometer sensors** for navigation and altitude hold.
- Addition of **First Person View (FPV)** systems for real-time aerial monitoring.
- Use of **IoT modules** for telemetry and data transmission.
- Applications in **aerial mapping, surveillance, environmental monitoring, and disaster assessment**.

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# APPARATUS

| Components | Specifications | Function | Quantity |
|---|---|---|---|
| BLDC Motor | 1000KV outrunner type | Provides thrust to propel the aircraft | 1 |
| ESC (Electronic Speed Controller | 30A, compatible with 2–3S LiPo | Controls the speed of the BLDC motor based on RF input | 1 |
| Servo Motors | 9g Micro Servo, 180° rotation | Control ailerons, rudder, and elevator | 4 |
| RF Transmitter and Receiver Module | 6-channel 2.4GHz | Sends and receives control signals for motor and servos | 1 |
| LiPo Battery | 3S 11.1V 2200mAh | Powers the ESC and servos | 1 |
| Thermocol Airframe | Lightweight and durable | Acts as the main body and wings | As required |
| Propeller | 10x4.5 inch | Converts motor torque into thrust | 1 |
| Control Rods and Hinges | Steel or carbon rods | Transfers servo motion to control surface | As required |

# WORKING PRINCIPLE

## 1. BLDC Motor (Brushless DC Motor)

Working Principle:
A Brushless DC (BLDC) motor operates based on the principle of electromagnetic induction. It consists of a rotor made of permanent magnets and a stator made of electromagnets (windings). When the stator windings are energized in a specific sequence, a rotating magnetic field is generated. This field interacts with the rotor's magnetic poles, causing it to rotate.

Mechanism Used:
In a BLDC motor, electronic commutation replaces the mechanical commutator and brushes used in conventional DC motors.The Electronic Speed Controller (ESC) supplies the motor with three-phase AC power by switching current through the stator coils in a precise sequence.The timing of this switching determines the speed and torque of the motor.The motor's speed (RPM) is proportional to the applied voltage and the KV rating (RPM per volt).

Role in Making the Plane Fly:
The BLDC motor drives the propeller, converting electrical energy into mechanical thrust.The forward thrust pushes the airplane through the air.As air flows over the wings, lift is generated according to Bernoulli's principle, allowing the plane to take off and stay airborne.Controlling the motor speed via the transmitter's throttle stick determines how fast the plane moves and how much lift it generates.

## 2. Servo Motor

Working Principle:
A servo motor operates on Pulse Width Modulation (PWM) control. The servo receives a series of electrical pulses from the receiver — the width of the pulse (1–2 ms) determines the angle of rotation.
For example:
1 ms → 0° position
1.5 ms → 90° (neutral)
2 ms → 180° position

Mechanism Used:
Inside the servo motor are:
A DC motor
A gear train to increase torque and reduce speed
A potentiometer for position feedback
A control circuit that compares input signal position to the actual shaft position and adjusts accordingly
This creates a closed-loop control system, maintaining precise angular positions.

Role in Making the Plane Fly:
Each servo motor controls one of the three main control surfaces:
Aileron servos: control the roll or banking motion.
Elevator servo: controls pitch (nose up or down).

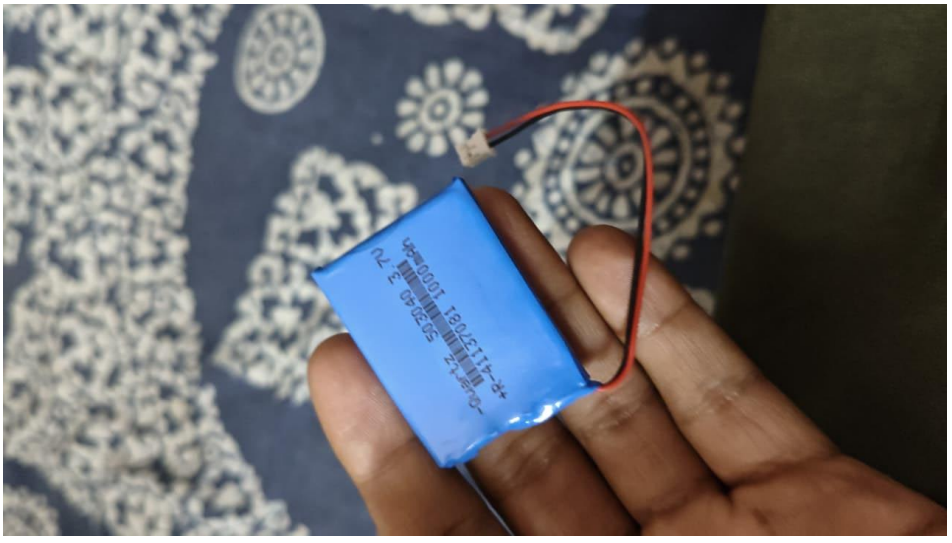
Rudder servo: controls yaw (left or right turn).
By changing the angles of these surfaces, servos directly affect the aircraft's orientation and stability in air.

### 3. Li-Po Battery (Lithium Polymer Battery)

Working Principle:
A Li-Po (Lithium-Polymer) battery uses lithium-ion electrochemistry. It consists of multiple cells, each producing about 3.7V nominal voltage. The battery releases energy through ion exchange between the positive (lithium cobalt oxide) and negative (graphite) electrodes via a polymer electrolyte.

Mechanism Used:
Provides direct current (DC) supply to all electronic components.A 3S battery (3 cells in series) gives 11.1V nominal and 12.6V fully charged voltage.High discharge rate (C-rating) ensures enough current is supplied to power the BLDC motor during flight.

Role in Making the Plane Fly:
zThe Li-Po battery is the main power source of the airplane. It:Supplies high current to the ESC which drives the BLDC motor.Powers the receiver and servos (through the ESC’s built-in BEC circuit).Without a lightweight and high-capacity battery like Li-Po, the airplane couldn’t sustain long flight durations or provide enough thrust.



**4. ESC (Electronic Speed Controller)**

Working Principle:
An ESC controls the speed of a BLDC motor by varying the duty cycle of electrical pulses (similar to AC switching). It uses Pulse Width Modulation (PWM) to regulate the amount of power delivered to the motor.

Mechanism Used:

Converts DC from Li-Po to 3-phase AC output.Uses MOSFETs (electronic switches) controlled by a microcontroller inside the ESC to generate alternating current.Receives a PWM signal (1–2 ms pulse width) from the receiver's throttle channel to determine motor speed.Contains a BEC (Battery Eliminator Circuit) that outputs 5V regulated power to the receiver and servos.

Role in Making the Plane Fly:
The ESC acts as the motor controller — it translates throttle input from the transmitter into motor speed.Low throttle = slower propeller speed → less thrust → descent.High throttle = faster propeller speed → more thrust → climb.
Thus, it controls the airplane's forward velocity and helps manage takeoff, cruising, and landing.

## 5. RF Receiver

Working Principle:
The receiver works on Radio Frequency communication. It receives encoded signals from the transmitter through a 2.4 GHz link and decodes them into Pulse Width Modulated (PWM) outputs.

Mechanism Used:
Each channel on the receiver corresponds to one control function (Throttle, Aileron, Elevator, Rudder).Uses AFHDS (Automatic Frequency Hopping Digital System) for interference-free communication.The decoded PWM signals are sent to the ESC and servo motors to actuate them accordingly.

Role in Making the Plane Fly:
The RF receiver acts as the control hub for the onboard electronics. It interprets the pilot’s joystick inputs and sends real-time control signals to:Servos (to move control surfaces)ESC (to vary motor speed)
This ensures smooth, synchronized movement and immediate response to commands during flight.

## 6. RF Transmitter

Working Principle:
An RF transmitter converts the pilot’s joystick movements into encoded digital signals and transmits them over a 2.4 GHz RF link.

Mechanism Used:

Uses AFHDS 2A protocol, which automatically switches between frequencies to prevent interference.The transmitter has multiple channels (typically 6) to send control data for throttle, ailerons, elevator, rudder, and auxiliary controls.Each control stick position is converted to a digital PWM signal which corresponds to a servo or ESC command.

Role in Making the Plane Fly:
The transmitter is the main control device used by the pilot.
It sends:Throttle signal → to ESC → controls BLDC motor speed.Aileron, Elevator, Rudder signals → to servos → control direction and stability.
This enables full manual control of the aircraft’s flight attitude and thrust from the ground.

**7. Control Surfaces**
A. Ailerons

Mechanism: Two hinged flaps on the trailing edge of each wing, controlled by separate servos.

Working Principle: When one aileron moves up and the other down, the lift on each wing changes — the airplane rolls (banks) around its longitudinal axis.

Role: Enables the plane to turn left or right by banking in the desired direction.

B. Elevator

Mechanism: Hinged flap on the horizontal stabilizer (tailplane), moved by one servo.

Working Principle:
Elevator up → tail down → nose rises → increases pitch → plane climbs.
Elevator down → tail up → nose lowers → decreases pitch → plane descends.
Role: Controls pitch movement (up and down orientation of the nose).

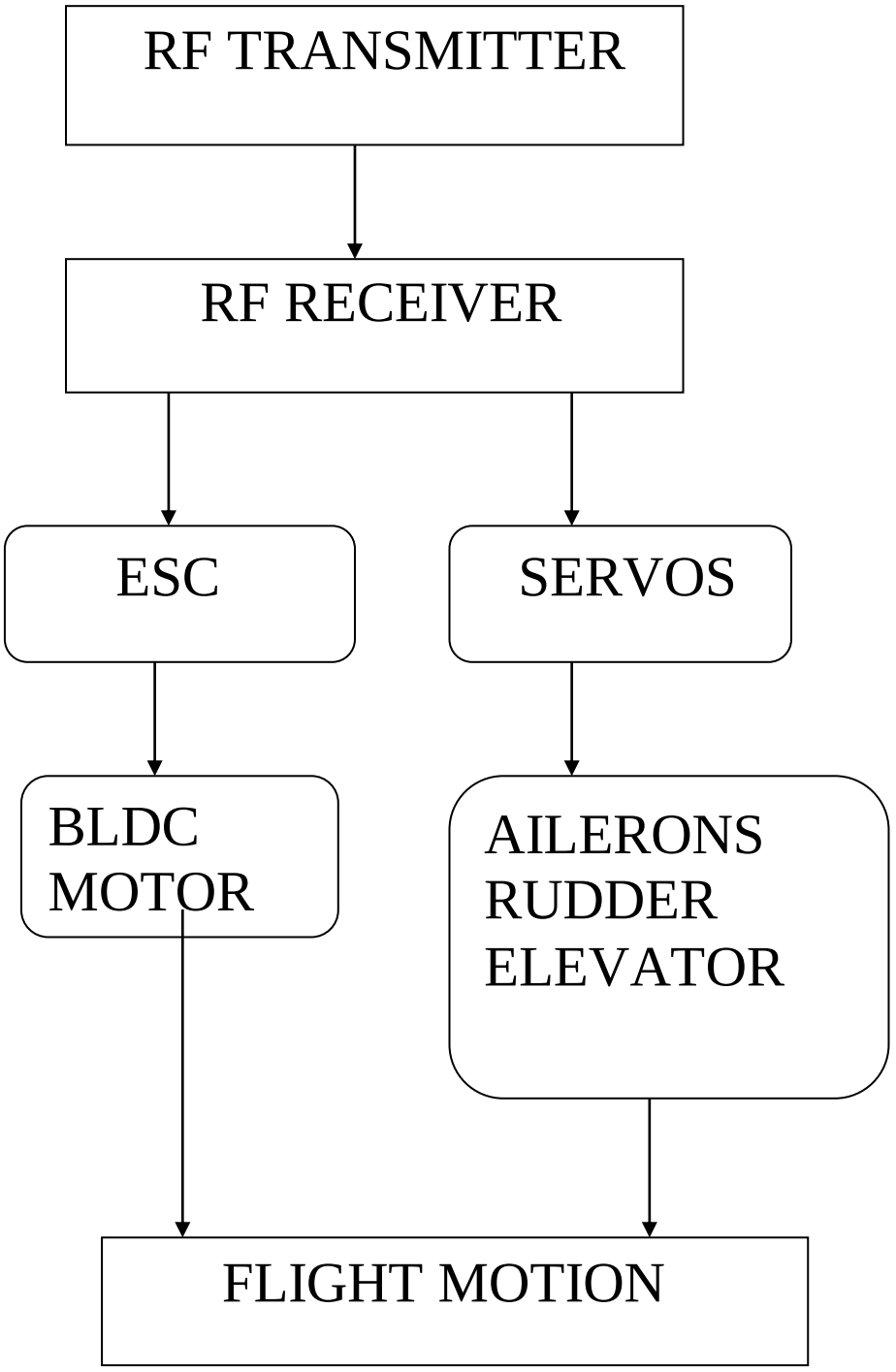
C. Rudder

Mechanism: Movable surface attached to the vertical stabilizer (fin), controlled by one servo.

Working Principle:
Deflecting the rudder changes the airflow on the tail → generates a sideways aerodynamic force → causes yaw (nose turns left/right).

Role: Controls yaw movement, assisting in smooth turns and directional stability.

**BLOCK DIAGRAM**



# CONCLUSION

The RC airplane project successfully demonstrates the integration of electronic control systems, aerodynamics, and communication technology to achieve stable and controlled flight. By combining components such as the BLDC motor, ESC, Li-Po battery, RF transmitter–receiver pair, and servo-controlled control surfaces (ailerons, rudder, and elevator), the model effectively simulates the functioning of a real aircraft. The transmitter-receiver system ensures smooth and real-time communication between the pilot and the aircraft, while the ESC and BLDC motor efficiently convert electrical energy into mechanical thrust. The servo motors provide precise control over flight orientation, ensuring stable maneuverability. Overall, this project showcases the practical application of electronics, control systems, and embedded hardware in aviation modeling and serves as a foundational step toward understanding modern unmanned aerial vehicles (UAVs).

# FUTURE PERSPECTIVE

In the future, the design can be enhanced by incorporating advanced technologies such as GPS-based navigation and autopilot systems for autonomous flight control. Integration of sensors like gyroscopes, accelerometers, and barometers can improve flight stability and allow for real-time telemetry feedback. Using a flight controller board such as an Arduino-based or Pixhawk system would enable the implementation of stabilization algorithms (PID control), altitude hold, and return-to-home functions. Further improvements can include using lightweight composite materials for better lift-to-weight ratio and extending the communication range through digital 2.4 GHz telemetry modules. With such developments, the project can evolve from a manually operated RC model into a fully autonomous UAV platform with potential applications in surveillance, environmental monitoring, and aerial mapping.

1.